Chandu Vemula

ca4443700@gmail.com | 9346281421 | Chandu Portfolio | github.com/IamChandu114 | linkedin.com/in/pace1304

Skills

Languages: Python, C++, JavaScript (ES6+), SQL.

AI / Machine Learning: PyTorch, TensorFlow, scikit-learn, Deep Learning, NumPy, Pandas, OpenCV, Computer Vision, Natural Language Processing (NLP), Multimodal Learning.

LLM & Generative AI: Hugging Face Transformers, LangChain, Large Language Models (LLMs), AI Agents.

Systems & Backend: Linux, FastAPI, Node.js, Express.js, REST APIs, Redis, Multithreading & Concurrency.

Cloud & DevOps: Docker, AWS (EC2, S3), Git, GitHub, CI/CD, Postman.

Databases: PostgreSQL, MySQL, MongoDB.

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, System Design.

Work Experience

Open-Source Contributor **Oct 2026 – Present**

vLLM | *Python, PyTorch, OpenCV*

- Authored PR #61096 (under review) addressing issue #60985, implementing early video-frame resizing for Qwen3-VL that reduced frame-buffer memory by 84% (13.18 → 2.11 MiB) in a synthetic 720p benchmark.
- Implemented processor-derived resize-budget propagation for Qwen3-VL and Cosmos3-Edge, with guarded early resizing and idempotency checks designed to preserve model-input equivalence.
- Added regression tests and a benchmark script; reported 7 targeted tests passing locally and addressed nested videos_kwargs sizing overrides following maintainer feedback, with regression coverage.

PRAM EduTech, Hyderabad **Oct 2026 – Dec 2026**

Software Engineer Intern

- Analyzed REST API request/response flows and backend data handling using Postman, validating payload structures, response behavior, data consistency across 20+ API endpoints in integrated application modules.
- Designed and executed functional, integration, and regression test scenarios across 10+ mobile application workflows, identifying cross-module failures and verifying corrective fixes.
- Systematically reproduced and investigated 10+ defects, collaborating with developers to isolate functional inconsistencies, validate resolutions, and improve application stability across 3+ releases.

Project Work

- **ASTRA OS — Autonomous Runtime Intelligence Platform (2026):** GitHub
  - *Python* | *FastAPI* | *PyTorch* | *scikit-learn* | *React* | *WebSockets*
  - Engineered a closed-loop system that forecasts CPU, memory, and thermal pressure and runs guardrail-constrained optimizations with rollback. Architected 26 backend modules and 40+ REST/WebSocket endpoints with auth and rate limiting; 16-module pytest suite.
- **Quantum RISC — RISC-V Processor Simulation & Verification Platform (2026):** GitHub
  - *Python* | *SystemVerilog* | *FastAPI*
  - Implemented single-cycle and five-stage pipelined RV32I CPUs with hazard detection and forwarding; built a WebSocket-driven verification Studio with 22 testbenches. Selected by XERJ for inclusion of the QuantumRISC Studio integration in its public Corpus Hub as a reference implementation for AI coding agents, with attribution.
- **Wave Core X — 5G/6G Baseband Signal Engineering Platform (2026):** GitHub
  - *Python* | *FastAPI* | *NumPy* | *SciPy* | *React* | *TypeScript*
  - Implemented an end-to-end baseband pipeline: Hamming coding, BPSK–256-QAM, OFDM, fading channels, ZF/MMSE, BER/SER/EVM. Built FastAPI endpoints and a React/TypeScript UI, validated by 41 passing tests with seeded reproducibility.

Education

Gurunanak Institutions Hyderabad **June 2026**

B.Tech. in Computer Science and Engineering- (AI/ML) **CGPA: 9.2/10**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, DBMS, OOP, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing.

Awards and Certificates

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate.
- Excellence Award for Contributions to AI Systems and Applied Computing.
- International Youth Math Challenge (IYMC) — Qualified.

Technical Leadership & Contributions

- **AI Systems Engineering Lead:** Architected scalable AI systems and optimized model performance.